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Effects of clay, gum Arabic and hybrid binders on the properties of rice and coffee HUSK briquettes

Josephine Moriku Thomas Celestino [a,b], Peter Okidi Lating [b,*], Betty Nabuuma [c,**], Vianney Andrew Yiga [c]

[a] *School of Applied and Industrial Sciences, University of Juba, P. O. BOX 82, Juba, South Sudan*
[b] *Department of Electrical and Computer Engineering, Makerere University, P.O. BOX 7062, Kampala, Uganda*
[c] *Department of Mechanical Engineering, Makerere University, P.O. BOX 7062, Kampala, Uganda*

A R T I C L E I N F O

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A B S T R A C T

The goal of this study was to determine the effects of clay, gum Arabic and hybrid binders on the physico-thermal and mechanical property of bio-composite briquettes from carbonize rice husks and coffee husks. Combined biochars briquettes were developed under low pressure (≤ 7 Mpa). The physical properties, density and calorific value of the briquettes were determined using thermogravimetric analysis and bomb calorimeter, respectively. The thermal properties were determined using water boiling time protocols. The particulate matter of 2.5 was determined using filter based gravimetric while the Carbon Monoxide and Carbon Dioxide levels were determined by electrochemical cell and dispersive infrared sensors, respectively. The results showed that the type of binder used in the development of bio-composite briquettes significantly affected their physico-thermal, mechanical property and calorific values. Heating values for bio-composite briquettes developed with clay binder ranged from 17140 to 18336 kJ/kg, briquettes with gum Arabic binder the calorific values ranged from 18053 to 18665 kJ/kg and the heating values of the bio-composite briquettes developed with hybrid binder ranged from 17404 to 18232 kJ/kg. The results of this work indicate that bio-composite briquettes with different binders can be applied as fuel in domestic and industrial application, however briquettes with gum Arabic had higher calorific values compared to briquettes with clay and briquettes with hybrid binder. The results obtained for all parameters showed that briquettes with a hybrid binders had a better performance based on boiling time had reduced, had higher burning rate and compressive strength, low fuel consumption. Therefore, this study recommends the use of the bio-composite briquettes produced with a hybrid binder and gum Arabic binder.

1. Introduction

The numerous energy challenges facing the world include fossil fuel shortages, emissions from combustion, scarcity of hydropower electricity sources, low access to modern energy technologies and high grant investment costs for energy efficient systems [1]. Africa faces significant energy challenges. Over 80 % of sub-Saharan Africa rely on biomass for cooking purposes [2]. In South Sudan, 86 % of the population uses firewood, and 10 % uses charcoal as their primary cooking fuel [3]. This has led to deforestation, soil erosion, carbon monoxide production, carbon dioxide, global warming, and general health implications [4]. Biomass is a sustainable and renewable energy source developed from agricultural residues, energy crops, animal manure, agro-industrial and food-processing residue, municipal solid wastes (MWS), wood wastes, aquatic plants, algae, sewage, and other biological resources [5]. The current exploitation of these renewable energy sources is highly inefficient. It causes climate change or global warming [6]. Climate change is a major threat to the socio-economic development of South Sudan which is one of the most rapidly warming places on earth, with temperatures increasing by 0.4 °C per decade. It is two and a half times higher than the global average [7,7,8]. More efficient methods of managing these energy materials include; gasification (gas for transportation), pyrolysis (biofuel for transportation and heating), and solid fuels (briquettes for heat). Gasification and biofuel have high investment cost compared to briquettes [9]. A briquette is a compressed block of coal dust or other combustible biomass materials used for fuel and kindling to start a fire.

* Corresponding author.
** Corresponding author.
E-mail addresses: peter.lating@mak.ac.ug (P.O. Lating), nabuuma@cedat.mak.ac.ug (B. Nabuuma).

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Advantages of using briquettes for energy production include: waste management which when these agricultural waste products are burned in open fields, or allowed to rot it presents an environmental concern, green energy, cheapness, and the ease of production [10,11].

There are three ways of producing briquettes: (i) high-pressure compaction which does not require binder, (ii) medium compaction method that requires medium pressure and binder, as well as (iii) a low pressure method which requires incorporation of a binder [11]. Binders have been used in the briquetting of different biomass materials to tightly hold the atoms together. They also have significant effects on thermal characteristics of the briquettes [12]. Binders are mainly divided into two types: organic and inorganic binders. The examples of organic binder include starch, molasses, while cement and clay are examples of inorganic binder [13]. Previous study used clay as a binder. It was observed that clay binder increase the bulk density and compressive strength of briquettes. Another study showed that the optimum briquette produced at 15 % clay had high calorific value. The calorific values decreased with increased of clay. Briquette produced from corn cobs, tender coconut husks and palm kernel shells with clay showed the decrease in calorific value compared to briquettes without clay binder [14–16]. Briquettes produced from water melon peel with starch and gum Arabic binders. The briquettes with starch binder had higher calorific value and took shorter time to boil water compared to briquettes from water melon peels with gum Arabic binder. Briquettes produced from sawdust with two binders - gum Arabic and starch. The combustion and physical properties of briquettes with starch binder performed better than briquettes with gum Arabic [17,18]. Molasses was used as a binder in production of briquettes from coconut shells and bambara nut shells. The results showed that the briquettes of mixed biomass had higher energy content compared to briquettes from individual biomass it was noted that both physical and combustion properties were significantly improved when the residues were mixed together [19]. In the study of briquettes produced from rice straw using sawdust as a binder showed that the addition of sawdust with rice straw improved briquettes stable density and shatter index. It also reduced the ash content and reduced compaction pressure requirement [20]. Neem powder was utilized as a binder in the production of briquette from sawdust it was found that briquettes had higher strength, handling, water resistance properties and lower calorific value than the sawdust briquettes [21]. Pulp paper and cow dung were used as a binder in the production of briquettes from groundnut shells it was found that briquettes with pulp paper binder had higher calorific value compared to briquettes with cow dung [22]. In the study of corncob and oil palm trunk bark using water pulp paper under low pressure techniques it was been found that combined of corncob and oil palm trunk bark briquettes had better qualities than the corncob briquettes [23]. Cow dung, pulp paper and starch were used as binder with saw dust and rice husks to produced briquettes it was been found that the briquettes with had higher calorific values followed by briquettes with starch binder and cow dung, however briquettes with cow dung had higher ash content [24]. Briquettes produced from breadfruit shell with African pear (resin) as binder it was be reported that the calorific value, ignition time and fuel efficiency increase with increase in resin content while burning rate and specific fuel consumption decreased with increase of resin binder [25]. Microalgae as a binder was used with wood residues to produced pellets briquettes it was been found that microalgae increase the bulk density and compressive strength [26]. In the study of rice husks, corn cob and bagasse briquettes with microalgae, biosolids and starch as a binders it were found that briquettes with microalgae had positive effect on strength and density while briquettes with biosolids had improves the density but had negative impact in compressive strength. Briquettes with starch reduced the green and relaxed densities([27] amongst others.

Briquetting process faces lots of challenges ranging from production of briquettes with poor combustion properties, swelling of briquettes, and competition of binding materials with consumable food products such as starch. Additionally, starch possess a threat to food security and swelling briquettes [28]. It is therefore necessary to find other alternative binders such as hybrid binders which consist of clay and gum Arabic. Indeed South Sudan is richly endowed with Gum Arabic from acacia trees. It grows in Central, Western and Northern parts of the country. The country harvests about 50,000 tons annually according to the South Sudan Gum Arabic Federal Union (SSGAFU) [29]. Therefore in this study, use of clay, gum Arabic and a hybrid consisting of gum Arabica and clay were compared as binders for production of carbonized bio composite briquettes from rice and coffee husks.

2. Experimental

2.1. Materials

Rice husks were obtained from the western Uganda and coffee husks from Namanve industrial Area Kampala. The binding materials in the form of Gum Arabic were collected from South Sudan and the clay from Bwaise Kampala.

2.2. Methods

2.2.1. Carbonization process

50 kg of rice husks and 50 kg of coffee husks were sun - dried for 4 h in two days to remove the initial moisture content. A steel drum carbonizer with a capacity of 210 L was used as a carbonizer. The carbonizer was fed with 25 kg of dried rice/coffee husks match was used for ignition under controlled temperature of 300–350 °C for 9 h. A pyrometer was used for temperature measurement. At the end of each shift, the rice husks biochar was poured on the dried surface, then water was sprinkled on biochar. The carbonized rice and coffee husks were pounded by woody mortar and pestle, then sieved through mesh 2 mm sizes.

2.2.2. Briquette development

For binder preparation, one kg of Gum Arabic was immersed into 2 L of water were kept for 24 h to produce liquid Gum Arabic binder. 2.4 %, 3.6 %, 4.7 %, 5.8 % and 6.9 of clay binder; Bio-chars of 200 g portions were mixed with 2.4 %, 3.6 %, 4.7 %, 5.8 % and 6.9 of Gum Arabic binder; again bio-chars of 200 g were mixed with 4.7 %, 6.9 %, 9 %, 11.1 % and 13 % of hybrid binder. The clay binder, gum Arabic binder and hybrid binder were then mixed with the bio-chars and fed in manual briquette machine under low pressure (≤7 Mpa). It is related to the pressure used by other researchers to develop low cost briquettes [30–32].

2.2.3. Physical properties

The moisture content, volatile matter, ash content, fixed carbon of the developed bio-composite briquettes were determined using Eltra thermostep Thermogravimetric analyzer (TGA) version: Tga1.4.3.2a3, according to ASTM D7582-15 were calculated using Equations (1)–(4). Experiments were carried out from room temperature to 920 °C with a heating rate of 16 °C/min. High-purity compressed air oxygen and nitrogen 99.5% was used. Nitrogen gas was used as the purge gas for pyrolysis experimentation. The flow rate was maintained at 1 L/min and the sample masses averaged 1.1 g [31,33].

$$\text{Moisture content } (\%) = \frac{A-B}{A} \times 100 \tag{1}$$

$$\text{Volatile matter } (\%) = \frac{B-C}{B} \times 100 \tag{2}$$

$$\text{Ash content } (\%) = \frac{D}{B} \times 100 \tag{3}$$

$$\text{Fixed carbon} = 100 - (\text{Moisture content} + \text{volatile matter} + \text{ash content}) \tag{4}$$

Where, A is the initial mass of briquettes (g), B is mass (g) after drying at 105 °*C*, C is mass (g) after devolatilization, D is mass (g) of ash.

2.2.4. Density and compressive strength

The density of briquettes was calculated by dividing the mass of briquettes by the volume of briquettes. The mass of briquettes specimen were measured by electric weight machine while the volume of fuel was measured by taking the height and diameter of briquettes by means of Vernier calliper [34,35]. A universal testing machine (Model: Testometric 0300–02032) to measure the compressive strength of briquettes. The test method consists of applied a compressive axial load to briquettes was applied at a constant rate of 2.000 mm/min until failure occurred. The compressive strength of the specimen was calculated by dividing the maximum load attained during the test by the cross-section area of the specimen.

2.2.5. Calorific values

Oxygen Bomb Calorimeter version IKA*C 2000 was used for determination of the calorific values of the briquettes specimen. Approximately one gram of sample was placed inside nickel crucible and fired inside the bomb calorimeter using the ignition wire in present of oxygen [31,36].

2.2.6. Combustion properties and emissions concentration

This test were carried out according to Water boiling test protocol (WBTP 4.2.3). The WBTP was used to measure time taken to boil 5 L of water, the burning rate, specific fuel consumptions, CO, CO_2 and particulate matter ($PM_{2.5}$). There are two types of test namely cold start phase and simmer phase. In cold start phase, initially an aluminum pot containing 5 L of water was placed on stove under Laboratory Emission Monitoring System (LEMS). A thermocouple temperature sensor up to 250 °C was immured inside the pot to record the temperature of water. Filters based method was used to determine particulate matter ($PM_{2.5}$). Two filters were Pre-weigh and post-weight. Pre-weight filter was fixed into a gravimetric cyclone, after the 5 L of water reached local boiling point. After the boiling point the pre-weight filter becomes post-weight filter. For measuring the concentration of CO an electrochemical cell sensor was used (which was the part of LEMS sensor box). Non-dispersive infrared was used to measure the concentration of CO_2.

3. Results and discussion

3.1. Physical properties

3.1.1. Physical properties

The moisture content results for bio-composite briquettes developed with clay binder (CYB), gum Arabic binder (GAB) and hybrid binder (HBB) are shown in Table 1. The moisture content for briquettes ranged from 7.1 % to 9.5 %. It supports the results of [37,38] recommended that for better performance of briquettes, the moisture content should fall within 5%–20 %. In this research therefore, there will be no energy loss because of no water evaporation during the combustion at expanse of calorific value of fuel as reported by Ref. [39]. Effect of briquettes with CYB on moisture content, the moisture content decreased with increase of CYB while the moisture content of briquettes with gum Arabic increased with increase of GAB. This could be due to complex mixture of hydrophilic carbohydrate and hydrophobic protein components of gum Arabic [40]. Hybrid binders had a positive effect on reducing the moisture content compared to briquettes with gum Arabic and briquettes with clay binder. However, the values obtained in this research were less than the moisture content of briquettes made from biochars of coffee and rice husks (10.98–13.72 %) [31] this could be because starch binder was applied. Also in the study of physico-mechanical characteristics of rice husk briquettes using gum Arabic binder had the moisture content of (8.03–8.07 %) [41]. Influence of briquettes with CYB and briquettes with GAB on volatile matter, had the same trend increase with increase of binders.The volatile matter of briquettes developed with CYB had the lowest volatile matter compared to the briquettes developed with HBB and the briquettes developed with GAB which had the highest volatile matter. The lowest volatile matter from briquettes with clay binder could be attributed to the fact that clay is inorganic material which means the volatile matter content from clay binder contribution is very small [42]. Also low volatile matter indicated that the ignitability of briquettes will be reduced and the combustion will produce very little or no smoke [43].

The effect of briquettes with clay binder on ash content, ash increased with increase of CYB while the opposite trend on ash content of briquettes with GAB. Briquettes with clay binder had the highest ash content compared to briquettes with HBB while briquettes with GAB which had the lowest ash content. The higher ash content in briquettes with clay was because both rice husks and clay binder ash have silicon dioxide (SiO_2) [42,44].The high ash content reduces heating value which produce higher thermal resistance to heat transfer and may generate slag deposits that need expensive equipment maintenance [45,46]. Fixed carbon of briquettes produced with clay binder, the fixed carbon decreased with the increase of clay binder this results is similar to the study where clay binder was used [42]. The effect of briquettes with GAB on fixed carbon, decreased with increase of GAB.The general overview was that the fixed carbon of briquettes produced with gum Arabic binder had highest fixed carbon compared to briquettes produced with CYB, then briquettes produced with HBB had the lowest fixed carbon.

Table 1
Physical properties of the briquettes.

Types of binder	Binder (g)	Binder (%)	MC %	MC-SD	VC %	VC-SD	Ash %	Ash-SD	FC %	FC-SD
Clay	5	2.4	7.9	0.10	28.2	0.14	25.2	0.39	38.5	0.61
Clay	7.5	3.6	7.8	0.18	28.7	0.32	26.1	0.10	37.2	0.24
Clay	10	4.7	7.7	0.17	28.3	0.23	26.8	0.43	37.1	0.48
Clay	12.5	5.8	9.5	3.16	35.2	12.35	11.5	27.45	43.6	11.94
Clay	15	6.9	7.6	0.04	28.3	0.35	27.7	0.56	36.1	0.24
Gum Arabic	5	2.4	7.6	0.12	29.3	0.44	23.7	0.16	39.2	0.40
Gum Arabic	7.5	3.6	7.6	0.08	30.0	0.46	23.8	0.20	38.5	0.60
Gum Arabic	10	4.7	7.7	0.20	29.8	0.58	24.0	0.98	38.3	0.71
Gum Arabic	12.5	5.8	7.7	0.13	30.2	0.74	23.4	0.17	38.5	0.81
Gum Arabic	15	6.9	8.1	0.13	31.5	0.59	22.9	0.18	37.2	0.90
hybrid	10	4.7	7.1	0.10	29.3	0.47	27.1	2.60	36.2	2.06
hybrid	15	6.9	7.5	0.05	29.8	0.35	26.1	0.20	36.5	0.12
hybrid	20	9	7.3	0.19	29.8	0.51	25.9	0.65	36.9	0.40
hybrid	25	11.1	7.2	0.09	29.8	1.11	26.8	0.41	36.0	0.86
hybrid	30	13	7.4	0.32	30.4	0.98	26.8	0.91	35.3	0.48

*MC = moisture content, SD=Standard deviation, VC = volatile matter, FC= Fixed carbon.

3.1.2. Thermogravimetric analysis

The weight loss and derivative thermogravimetry (DTG), DTG is rate of weight change over time (%/min). Weight loss and DTG results for bio-composite briquettes developed with CYB, GAB and HBB are shown in Figs. 1–3. The weight loss behavior of briquettes results from an increase in temperature, from room temperature to 1000 °C. It was observed that the briquettes' weight loss remained at a plateau from combustion at room temperature (25 °C) to about 103 °C while the curved portion implied weight loss occurred from about 104 °C to about 105 °C. This weight loss occurred due to the evaporation of moisture and volatiles in the briquettes. The beginning of hemicellulose decomposition varies in the developed bio-composite briquettes from 263 °C to 386 °C. Between 600 °C and 758 °C, the decomposition of cellulose occurred along with the atmosphere's shift from nitrogen to oxygen. When the temperature rises above 800 °C, lignin decomposes, and the remaining weight percentage is mainly composed of residues, including ash tars and fixed carbon content [31].

3.2. Density and compressive strength

3.2.1. Density

The density results of bio-composite briquettes developed with a clay binder (CYB), Gum Arabic binder (GAB), hybrid binders (HBB), are shown in Table 2. The density for briquettes developed with clay binder had satisfied the purpose of clay reported to increase the density and hardness of briquettes [44]. The densities in this research correspond to the European 364 standards EN 14961-3 which stipulates briquette density to be > 0.90 g/cm^3 [47]. All the briquettes in this study, with different binders were observed to increase with increased of binders. This attributed to the fact that carbonization process reduce the volume of husks by increasing the density of the fuel.

3.2.2. Compressive strengths

Compressive strengths results for bio-composite briquettes developed with clay binder, Gum Arabic binder, hybrid binders, are shown in Table 3. All the compressive strengths of developed briquettes had increase with the increased of binders, this is in line with the briquettes developed from rice husks with gum Arabic binder [41]. The purpose of compressive strength is to indicate handle-ability, transportation and storage. It also shows the extent to which briquettes can be stacked on top of each other during domestic or kiln cooking [31].

3.3. Calorific values

3.3.1. Calorific values

The results for the heating values for bio-composite briquettes produced with clay binder, gum Arabic binder, hybrid binders, are shown in Table 4. Generally for briquettes with clay binder, heating value was observed to decrease with increase of CYB. This was in line with the study of briquettes produced from Sesame Stalk using clay binder [15]. The lowest heating value occurred due to high level of ash in briquettes. Clay binder in relation to starch binder, briquettes with starch binder from literature was found to have higher heating values compared to clay binder [10,32,48]. Briquettes with GAB had higher calorific values because briquettes with gum Arabic binder had lower ash content. Briquettes with CYB had lower calorific value, this because briquettes with CYB had highest ash content, this due to SiO_2 ash in clay binder [42,44]. However, the calorific values results attained in this research were higher compared to calorific value of briquettes made from rice husks and clay binder (9.47 MJ/kg) [49].

3.4. Combustion properties and emission concentration

3.4.1. Water boiling time

The results of water boiling test (WBT) for bio-composite briquettes developed with clay binder, Gum Arabic binder, hybrid binders are shown in Table 5. Briquettes with CYB took longer to boil 5 l of water compared to briquettes with GAB. This attributed to the fact that briquettes with GAB had higher calorific values and burning rates. Generally, it was been observed that the time to boil 5 l of water for all briquettes with different binders depends on the burning rate the higher the burning rate the shorter the time to boil water. This is similar to conclusion drown by author, the burning rate (how fast the fuel burns) and the calorific value (how much heat released) are the two factors that controlled the water boiling time [50]. However, the water boiling time results obtained in this research took shorter time compared to carbonized rice husks briquettes with gum Arabic binder took (18–22 min) to boil 1 L of water [51]. Briquettes produced from watermelon peels with gum Arabic took (24 min) to boil 1 L of water [17]. Development of rice husk briquettes with gum Arabic binder (15 min) to boil 1 L of water [52]. Also, in carbonized coffee husks briquettes with clay binder and carbonized rice husks briquettes with clay binder took (15–29 min)(15–44 min) respectively, to boil 1 L of water ([33].

3.4.2. Burning rates

Burning rates (BR) results for bio-composite briquettes developed from combined biochars with clay binder, Gum Arabic binder, hybrid binders are presented in Table 6. It was been observed that the BR for briquettes with CYB had the lowest BR, because clay is inorganic material. This was similar to comparative study of briquettes developed from carbonized sugar cane bagasse with clay binder had lower BR than briquettes with molasses binder [53]. However, the BR of briquettes with GAB had the highest BR, this is due to the fact that gum Arabic is combustible in nature. Generally, It was been observed that the higher volatile matter and low ash content had higher BR. The BR acquired in this research were higher compared to rice husks briquettes (0.001691 g/min) [54]. Also briquettes developed from rice husks and red bean skin (0.017 g/s) [55].Carbonized sugar cane bagasse with clay (0.32 g/min) [53]. Combustion performance of biomass composite briquette from rice husk and banana residue (0.1476 g/min) (0.2036 g/min)

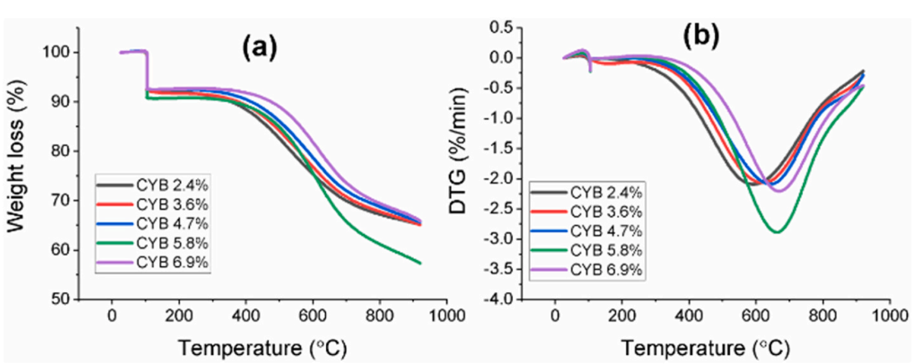

Fig. 1. Weight loss and DTG for bio-composite briquettes developed with CYB.

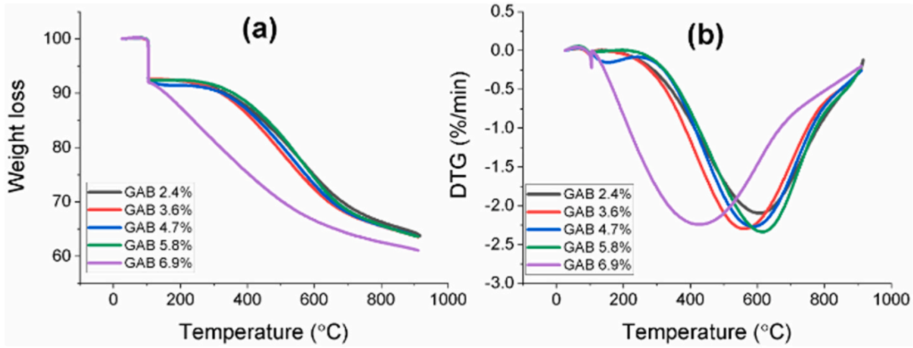

Fig. 2. Weight loss and DTG for bio-composite briquettes developed with GAB.

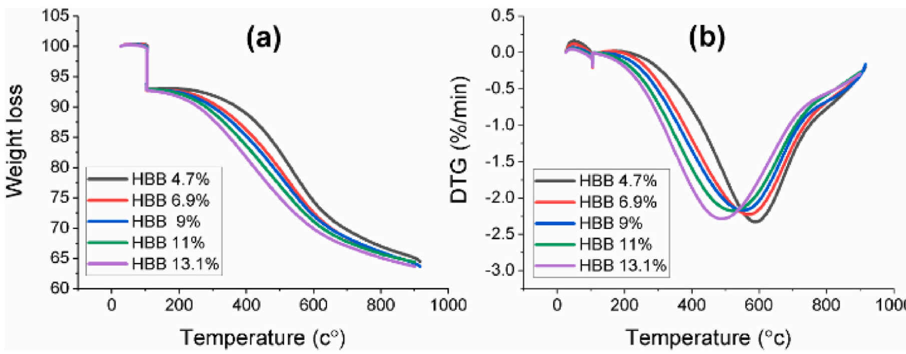

Fig. 3. Weight loss and DTG for bio-composite briquettes developed with HBB.

Table 2
Density of bio-composite briquettes.

Types of binder	Binder(g)	Binder (%)	Density (kg/m^3)
Clay binder	5	2.4	14.69
Clay binder	7.5	3.6	14.69
Clay binder	10	4.7	15
Clay binder	12.5	5.8	16.55
Clay binder	15	6.9	15.81
Gum Arabic binder	5	2.4	14.38
Gum Arabic binder	7.5	3.6	14.69
Gum Arabic binder	10	4.7	15
Gum Arabic binder	12.5	5.8	15
Gum Arabic binder	15	6.9	15.81
hybrid binders	10	4.7	14.38
hybrid binders	15	6.9	16.21
hybrid binders	20	9	15
hybrid binders	25	11.1	15.483
hybrid binders	30	13	17.5

Table 3
Compressive strength of bio-composite briquettes.

Types of binder	Binder(g)	Binder (%)	Compressive strength (kN/m^2)
Clay binder	5	2.4	1.62
Clay binder	7.5	3.6	3.18
Clay binder	10	4.7	2.43
Clay binder	12.5	5.8	3.63
Clay binder	15	6.9	5.13
Gum Arabic binder	5	2.4	2.52
Gum Arabic binder	7.5	3.6	3.87
Gum Arabic binder	10	4.7	2.83
Gum Arabic binder	12.5	5.8	5.24
Gum Arabic binder	15	6.9	8.3
hybrid binders	10	4.7	4.38
hybrid binders	15	6.9	6.45
hybrid binders	20	9	6.21
hybrid binders	25	11.1	11.75
hybrid binders	30	13	12.25

(0.2055 g/min) [56].

3.4.3. *Specific fuel consumptions*

The results of the specific fuel consumptions (SFC) for bio-composite briquettes developed with clay binder, Gum Arabic binder, hybrid binders, are shown in Table 7. Briquettes with CYB had the highest SFC, followed by briquettes with GAB then briquettes with HBB. Generally, the SFC of briquettes in this research, it depends on the calorific value and the density of fuel. The higher the calorific value and density the lower the SFC. Also, when the burning rate is low, it implies that the time to boil water is prolonged and it consumes a lot of fuel. However, briquettes with CYB had higher SFC compared to briquettes with GAB. It indicates that briquettes with clay and briquettes with GAB can be used in both low and higher cooking particularity in bakery for heat retention. SFC results achieved in this research were higher compared to composed briquettes which had SFC of (2.92–3.27 g/l) [57].While SFC results for this research were lower, compared to breadfruit shell briquettes, SFC range from (92–178 g/l) [25]. Also, palm kernel shell biochar briquettes had a SFC of (21 g/l) [58].

3.4.4. *Total emissions*

The results of the total emission of bio-composite briquettes produced with clay binder, Gum Arabic binder, hybrid binders are shown in Table 8. The results of the total emission of briquettes with CYB at cold start phase CO and CO_2 increased with increase of CYB up to 3.6 %.It then decreased with an increase of CYB however PM2.5 reduced with an increase of binder up to 4.7 %. It then increased with an increase of CYB exceptions of 6.9 %. At simmer phase CO, CO_2 and $PM_{2.5}$ had reduce compared to cold start. Furthermore, the results of the total emission of briquettes with GAB, at cold start CO increased with an increase of GAB exclusive of 4.7 % and 6.9 % which had similar values, while CO_2 reduced with an increase of GAB up to 3.6 % and increased with an

Table 4
Calorific values of bio-composite briquettes.

Types of binder	Binder(g)	Binder (%)	Calorific values (kJ/kg)
Clay binder	5	2.4	18336
Clay binder	7.5	3.6	18265
Clay binder	10	4.7	17517
Clay binder	12.5	5.8	17156
Clay binder	15	6.9	17140
Gum Arabic binder	5	2.4	18522
Gum Arabic binder	7.5	3.6	18053
Gum Arabic binder	10	4.7	18465
Gum Arabic binder	12.5	5.8	18102
Gum Arabic binder	15	6.9	18665
hybrid binders	10	4.7	17465
hybrid binders	15	6.9	18232
hybrid binders	20	9	17765
hybrid binders	25	11.1	17732
hybrid binders	30	13	17404

Table 5
Water boiling time of bio-composite briquettes.

Types of binder	Binder(g)	Binder (%)	Water boiling time (min)
Clay binder	5	2.4	74
Clay binder	7.5	3.6	71
Clay binder	10	4.7	57
Clay binder	12.5	5.8	45
Clay binder	15	6.9	69
Gum Arabic binder	5	2.4	42
Gum Arabic binder	7.5	3.6	43
Gum Arabic binder	10	4.7	50
Gum Arabic binder	12.5	5.8	46
Gum Arabic binder	15	6.9	47
hybrid binders	10	4.7	40
hybrid binders	15	6.9	41
hybrid binders	20	9	49
hybrid binders	25	11.1	48
hybrid binders	30	13	56

Table 6
Burning rate of bio-composite briquettes.

Types of binder	Binder(g)	Binder (%)	Burning rate (g/min)
Clay binder	5	2.4	2.88
Clay binder	7.5	3.6	3.04
Clay binder	10	4.7	3.77
Clay binder	12.5	5.8	4.47
Clay binder	15	6.9	3.04
Gum Arabic binder	5	2.4	4.55
Gum Arabic binder	7.5	3.6	4.93
Gum Arabic binder	10	4.7	3.89
Gum Arabic binder	12.5	5.8	4.73
Gum Arabic binder	15	6.9	4.16
hybrid binders	10	4.7	5.12
hybrid binders	15	6.9	4.64
hybrid binders	20	9	4.07
hybrid binders	25	11.1	4.23
hybrid binders	30	13	3.59

increase of GAB. While $PM_{2.5}$ increased with an increase of GAB up to 3.6 % then decreased with an increase of GAB. At a simmer phase the CO, CO_2 and $PM_{2.5}$ decreased compared to a cold start. The results of the total emission of briquettes with HBB at the cold start CO and CO_2 decreased with the increase of HBB exceptions of 13 %. While $PM_{2.5}$ increased with the increase of binder up to 6.9 %, it then decreased with an increase of binder. At simmer phase the CO, CO_2 and $PM_{2.5}$ had decreased rapidly compared a cold start. Furthermore, $PM_{2.5}$ was higher when both heating value and fixed carbon were higher, in all briquettes with CYB, GAB and HBB. This observation was opposite to research done on the characterization of $PM_{2.5}$ and gaseous emissions during combustion of ultra-clean biomass via dual-stage treatment [59]. However, for all briquettes it was been observed that higher volatile matter and lower fixed carbon resulted in lower $PM_{2.5}$.

Table 7
Specific Fuel Consumption of bio-composite briquettes.

Types of binder	Binder(g)	Binder (%)	Specific fuel consumption (G/L)
Clay binder	5	2.4	43
Clay binder	7.5	3.6	44
Clay binder	10	4.7	43
Clay binder	12.5	5.8	41
Clay binder	15	6.9	42
Gum Arabic binder	5	2.4	38
Gum Arabic binder	7.5	3.6	43
Gum Arabic binder	10	4.7	39
Gum Arabic binder	12.5	5.8	44
Gum Arabic binder	15	6.9	39
hybrid binders	10	4.7	41
hybrid binders	15	6.9	38
hybrid binders	20	9	40
hybrid binders	25	11.1	41
hybrid binders	30	13	40

3.4.5. Specific emission rate

Specific emission rate results of bio-composite briquettes developed with clay binder, Gum Arabic binder, hybrid binders, are presented in Table 9. The briquettes with a clay binder, CO of cold start increased with the increase of binder up to 3.6 % then remain constant. While CO2 increased with the increase of binder except for 4.7 % and 6.9 %. On the other hand $PM_{2.5,}$ had constant values first then decreased with the increase of binder exceptions of 5.8 %. At simmer phase, there was decreased compared to cold start phase CO_2, $PM_{2.5}$.The CO exceptions of 2.4 % which remain the same as cold start conditions. Furthermore, the specific emission rate of briquettes developed from combined biochars with GAB at the cold start. The values of CO was constant up to 3.6 % and then deceased with the increase of GAB exceptions of 5.8 %. However, the CO_2 decreased with the increase of binder from 3.6 % and increased with the increased of GAB. Also, PM2.5 increased with an increase of GAB binder up 3.6 % and decreased with the increase of binder excluding 5.8 %. At simmer phase there was a decrease in CO_2, $PM_{2.5}$ compared to a cold start. The CO decreased exceptions of 6.9 % and remained the same as at cold start.

The results of the specific emission rate of briquettes developed with the HBB, at the cold start CO had a constant values up 9 % and it decreased from the constant values with the increase of the binder. The CO_2 and $PM_{2.5}$ decreased with an increase of HBB. At simmer phase CO, CO_2 and $PM_{2.5}$ decreased compared to a cold start.

3.4.6. Emission rate

The emission rate of bio-composite briquettes with clay binder, Gum Arabic, are shown in Table 10. The emission rate results of briquettes developed with CYB at cold start CO and CO_2 increased with increased of binder exceptions of 6.9 % while $PM_{2.5}$ had constant values then decreased at 4.7 %, 6.9 % and increased rapidly at 5.8 %. At simmer phase CO and CO_2 decreased with an increase of the binder while $PM_{2.5}$ increased with an increase of binder up to 3.6 % then had constant value with an increase of binder except for 6.9 %.

The emission rate results of the briquettes developed with GAB at the cold start CO had constant value up 3.6 %. It then decreased with an increase of binder except for 5.8 %. The CO_2 decreased with increased of binder up to 3.6 % than it increased with the increase of the binder while $PM_{2.5}$ increased with an increase of binder up to 3.6 % then decreased with an increase of binder exceptions of 5.8 %. At simmer phase CO had constant values up to 4.7 % and increased with increased of binder which was constant values again while CO_2 increased with an increase of binder up to 3.6 % then decreased with an increase of binder exceptions of 6.9 %. However, $PM_{2.5}$ increased with increased of binder up to 3.6 % the decreased with increased of binder except 6.9 %.

Table 8
Total emissions of bio-composite briquettes with clay binder.

Cold start					Simmering		
Types of binder	Binder (%)	CO (g)	CO_2 (g)	$PM_{2.5}$ (mg)	CO (g)	CO_2 (g)	$PM_{2.5}$(mg)
Clay binder	2.4	50.13	222	1045	12.98	75	126
Clay binder	3.6	56.59	228	1005	7.78	74	177
Clay binder	4.7	52.47	175	702	4.57	60	144
Clay binder	5.8	38.86	188	1092	8.16	59	143
Clay binder	6.9	52.01	128	916	13.81	20	174
Gum Arabic binder	2.4	36.39	200	1073	9.40	67	110
Gum Arabic binder	3.6	40.58	48	1500	7.78	180	164
Gum Arabic binder	4.7	34.20	181	1065	9.52	50	122
Gum Arabic binder	5.8	66.68	215	1442	13.72	46	264
Gum Arabic binder	6.9	33.04	229	1037	11.75	105	192
hybrid binders	4.7	45.12	218	1337	4.79	55	76
hybrid binders	6.9	43.17	187	1385	6.19	55	179
hybrid binders	9	43.48	165	1229	1.96	52	214
hybrid binders	11.1	32.57	98	1263	7.78	74	134
hybrid binders	13	38.78	176	599	4.40	56	216

Table 9
Specific emission rate of bio-composite briquettes with CYB.

Cold start					Simmering		
Types of binder	Binder (%)	CO (g/min/L)	CO_2 (g/min/L)	$PM_{2.5}$ (mg/min/L)	CO (g/min/L)	CO_2 (g/min/L)	$PM_{2.5}$ (mg/min/L)
Clay binder	2.4	0.1	0.6	2.9	0.1	0.4	0.6
Clay binder	3.6	0.2	0.7	2.9	0.04	0.4	0.8
Clay binder	4.7	0.2	0.6	2.5	0.02	0.3	0.7
Clay binder	5.8	0.2	0.8	4.9	0.04	0.3	0.7
Clay binder	6.9	0.2	0.4	2.7	0.1	0.1	0.8
Gum Arabic binder	2.4	0.2	1	5.2	0.04	0.3	0.5
Gum Arabic binder	3.6	0.2	0.2	7.1	0.04	0.9	0.8
Gum Arabic binder	4.7	0.1	0.7	4.3	0.05	0.2	0.6
Gum Arabic binder	5.8	0.3	0.9	6.3	0.1	0.2	1.2
Gum Arabic binder	6.9	0.1	1	4.6	0.1	0.5	0.9
hybrid binders	4.7	0.2	1.1	6.8	0.02	0.3	0.4
hybrid binders	6.9	0.2	0.9	6.9	0.03	0.3	0.8
hybrid binders	9	0.2	0.7	5.1	0.01	0.3	1.3
hybrid binders	11.1	0.1	0.4	5.4	0.04	0.4	0.6
hybrid binders	13	0.1	0.6	2.2	0.02	0.3	1.1

Table 10
Emission rate of bio-composite briquettes with clay binder.

Cold start					Simmering		
Types of binder	Binder (%)	CO (g/min)	CO_2 (g/min)	$PM_{2.5}$ (mg/min)	CO (g/min)	CO_2 (g/min)	$PM_{2.5}$ (mg/min)
Clay binder	2.4	0.7	3.0	14.1	0.3	1.7	2.8
Clay binder	3.6	0.8	3.2	14.1	0.2	1.6	3.9
Clay binder	4.7	0.9	3.1	12.4	0.1	1.3	3.2
Clay binder	5.8	0.9	4.1	24.0	0.2	1.3	3.2
Clay binder	6.9	0.8	1.9	13.3	0.3	0.4	3.9
Gum Arabic binder	2.4	0.9	4.8	25.7	0.2	1.5	2.5
Gum Arabic binder	3.6	0.9	1.1	34.6	0.2	4.0	3.6
Gum Arabic binder	4.7	0.7	3.6	21.2	0.2	1.1	2.7
Gum Arabic binder	5.8	1.4	4.6	31.1	0.3	1.0	5.9
Gum Arabic binder	6.9	0.7	4.9	22.2	0.3	2.3	4.3
hybrid binders	4.7	1.1	5.5	33.6	0.1	1.2	1.7
hybrid binders	6.9	1.1	4.6	34.1	0.1	1.2	4.0
hybrid binders	9	0.9	3.4	25.1	0.1	1.5	6.1
hybrid binders	11.1	0.7	2.0	26.5	0.2	1.6	3.0
hybrid binders	13	0.7	3.2	10.8	0.1	1.4	5.5

Emission rate results of briquettes developed with HBB CO at the cold start had constant value up to 6.9 % then decreased with increase of the binder. While CO_2 decreased with increase of binder exceptions of 13 %. On the other hand, the PM $_{2.5}$ increased with increase of HBB up to 6.9 % and decreased with an increase of binder exceptions of 11.1 %. At simmer phase CO had constant values. It decreased at cold stare phase except for 11.1 %. The CO_2 increased with increase of binder except for 13 %. It decreased compared to cold stare, PM $_{2.5}$ cold start.

4. Conclusions and Recommendations

4.1. Conclusions

I. The study found out that the physical properties of the bio-composite briquettes with three different binders: clay binder, Gum Arabic binder and hybrid binders. The moisture content were in line with the recommended moisture content for briquettes. The bio-composite briquettes produced with clay binder had the lowest volatile matter followed by briquettes with hybrid binders, then briquettes with gum Arabic binder had the highest volatile matter. The ash content of briquettes developed with gum Arabic binder had the lowest ash followed by briquettes with hybrid binders, then briquettes with clay binder. The fixed carbon of briquettes developed with gum Arabic binder had the highest fixed carbon followed by briquettes with clay binder, then briquettes developed with hybrid binders had the lowest fixed carbon.

II. The heating values of bio-composite briquettes with gum Arabic binder had the highest calorific values, while briquettes produced with clay binder had the lowest heating values. However, the briquettes with hybrid binders had the highest density, followed by briquettes with clay binder, while briquettes with gum Arabic binder had lowest density. The compressive strength of briquettes with hybrid binders had the highest compressive strengths, followed by briquettes with gum Arabic binder, then briquettes with clay binder had the lowest compressive strengths.

III. Briquettes developed with CYB to boiling 5 L of water took the longest minutes, followed by briquettes with GAB, while briquettes with HBB had the shortest time. The higher the burning rate the shorter the time to boil water. The burning rate of briquettes with hybrid binders had higher burning rates followed by briquettes with gum Arabic binder then briquettes with clay binder had the lowest burning rate.

IV. The specific fuel consumptions of briquettes with hybrid binders were lower than briquettes with gum Arabic binder and clay binder. The fuel consumed depended on the heating value of the fuel. The burning rate and the volatile matter. When the volatile matter was high and the burning rate was less than 4.7 (g/min) the fuel consumed was smaller, which was good economically. When the burning rate was low, it implies that the time to boil was prolonged and specific fuel consumption was high.

V. From total emission, it was been found that the carbon monoxide, carbon dioxide and particulate matter did not exceed the standard range.

4.2. Recommendations

- All the briquettes industries and small to medium enterprises that apply medium pressure compaction method or low pressure compaction method which requires binder, this research is highly recommending the implementation of Gum Arabic as a binder for the food security purposes.
- Policy makers, in line with SDGs goal number two zero hunger the use of gum Arabic binder will reduce on the use of starch as binder. Clean energy which is goal number seven. South Sudan government should put more efforts on renewable energy sector.

4.3. Further research

This research was experimental in nature, the research determined the physical property, calorific value, briquettes density and compressive strengths of the briquettes. Water boiling test protocol was undertaken to determine water boiling time, burning rate, specific fuel consumption and CO, CO_2 and $PM_{2.5}$. In the thermogravimetric analysis, this research had considered proximate analysis only. The water boiling test protocol with target of emission, research consider CO, CO_2 and $PM_{2.5}$ without considering Sulfur dioxide and other harmful gases. Therefore, future research could be in improving on the calorific value of briquettes by maybe increasing on the concentration ratio of hybrid binders and gum Arabic binder, this should involve putting the ultimate analysis and other harmful gases released during combustion of fuel/briquettes into considerations. Also the cost analysis should be considered and the constituents of the clay and gum Arabic need to be examine.

Declaration of competing interest

I am pleased to submit a manuscript entitled "Effects Of Clay, Gum Arabic And Hybrid Binders On The Properties of Rice And Coffee Husk Briquettes" by Josephine Moriku Thomas Celestino, Peter Okidi Lating, Betty Nabuuma, Vianney Andrew Yiga. For consideration for publication in Result in Enginnering.We confirm that we have removed any identifying content that could compromise a blind review. This manuscript has not been published and is not under consideration for publication elsewhere. We have no conflicts of interest to disclose.

Data availability

Data will be made available on request.

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